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Recitation : 8

Decision Tree

Suspicious words | Money Keywords | Sender Known | Attachments | Class

ID3 : Entropy & Info Gain algorithm

* Calculate Info gain of Each feature

IG → max → best feature

min Entropy, max Info Gain [IG]

$$\text{Entropy} = \sum_{i=1}^{N} = p_i * \log_2(p_i)$$

$$\log_2(p_i) = \frac{\log_{10} p_i}{\log_{10} 2} = \frac{\ln p_i}{\ln 2}$$

$$\Rightarrow \text{Entropy (Parent)}$$

$$IG = \left(- \sum_{K} \frac{N_K}{N} = \text{Entropy}(K)\right)$$

1) $p_{Spam} = \frac{5}{10} = 0.5$

2) $p_{Non\ spam} = 0.5$

$E = 0.5 + 0.5 = 1$

1) Entropy (Sender Known)

1 -> Spam (1)
-> Non Spam (5)

0 -> Spam (4)
-> Non Spam (0)

2) Entropy (Many Key words)

Entropy = 0 Perfectly = 0

$$-\left[\frac{1}{6}\log_2\left(\frac{1}{6}\right)\right] + \left[\frac{5}{6}\log_2\left(\frac{5}{6}\right)\right]$$

$$+ 0 + 1(\log_2 1)$$

$$\Rightarrow -\left[-0.43 + (-0.219)\right]$$

$$\Rightarrow 0.65$$

$$IG = 1 - \left(\frac{6}{10} \times 0.65 + \frac{4}{10} \times 0\right)$$

$$\Rightarrow 0.61$$

Entropy (Has Attachment)

0 -> Spam (2)
-> N Spam (3) } Entropy > 0

2 $1 \rightarrow$ Spam (3)
$\rightarrow$ N Spam (2) } Entropy > 0

$IG < 1$

$\Rightarrow$ Entropy (Suspicious words)

4) Entropy > 0

$IG < 1$

Root node = Money keywords

because (0) = Non spam
(1) = spam

depth = 1, as there only final decision perfectly

3) Weighted Error

$$\varepsilon_t = \sum_{i=1}^{n} w_i \cdot I\left(y_i \neq h_i(x_i)\right)$$

becomes 1
if equal becomes 0

$$= \left(0.2\times 0 + 0.2\times 0 + 0.2\times 0 + 0.2\times 0 + 0.2\times 0\right)$$

$= 0$ No Error
'cause perfectly classified

4) Classifier Error / model Error

$$\alpha_t = \frac{1}{2} \ln\left(\frac{1-\varepsilon_t}{\varepsilon_t}\right)$$

$$= \frac{1}{2} \ln\left(1-0/0\right) = \infty$$

$$= 1/2 \ln(0) = \infty$$

5) Updated weight

$$w_i^{New} = w_i \cdot e^{-y_i \alpha_i h_i(x_i)}$$

$$= e^{-\infty} = 0$$

'Cause it is perfectly classified

6) Final Prediction, $H(x) = h_i(x_i)$